

Lesson 11 Homework - Calculus

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1

We want to find the first and second order derivatives of

$$r = \frac{(\theta - 1)(\theta^2 + \theta + 1)}{\theta^3}$$

We can see that the numerator is the factorization of difference of cubes, hence we have

$$r = \frac{\theta^3 - 1}{\theta} = 1 + \frac{1}{\theta^3} = 1 + \theta^{-3}$$

Then differentiating using the power rule, we have

$$\frac{dr}{dt} = 1 - 3\theta^{-4} = 1 - \frac{3}{\theta^4}$$

Then the second order derivative is

$$\frac{d^2r}{dt^2} = \frac{d}{dt}(1 - 3\theta^{-4}) = 1 + \frac{12}{\theta^5}$$

2

Since it passes through $(1, 2)$, we have

$$a + b + c = 2$$

And since it passes through $(0, 0)$,

$$c = 0$$

Since we know that $y = ax^2 + bx + c$ is tangent to $y = x$ at the origin, we can find the derivative of y

$$\frac{dy}{dx} = 2ax + b$$

when we substitute in $(0, 0)$ has to have equal 1 since it has the same slope as $y = x$. Hence,

$$y'(0) = b = 1$$

Then plugging in $(1, 2)$, with $b = 1, c = 0$, we get

$$a + 1 + 0 = 2 \implies a = 1$$

Hence

$$y = x^2 + x$$

3

3.1

We want to find the derivative of

$$y = 3 \cosh x + 7^x$$

which we can write as from the definition of $\cosh$

$$y = 3 \cdot \frac{e^x + e^{-x}}{2} + 7^x$$

we can differentiate each term and then add them together.

$$\frac{d}{dx} \frac{3}{2} (e^x + e^{-x}) = \frac{3}{2} (e^x - e^{-x}) = 3 \sinh x$$

$$\frac{d}{dx} 7^x = 7^x \ln(7)$$

Hence

$$\frac{dy}{dx} = 3 \sinh x + 7^x \ln(7)$$

3.2

We want to find the derivative of

$$f(x) = \sinh x + \cosh x - 2x$$

By deriving each term

$$\frac{d}{dx} \sinh x = \cosh x$$

$$\frac{d}{dx} \cosh x = \sinh x$$

$$\frac{d}{dx} 2x = 2$$

Hence,

$$\frac{df}{dx} = \sinh x + \cosh x + 2$$

3.3

We want to find the derivative of

$$g(u) = u \cosh^2(u) + \pi^u - u \sinh^2(u)$$

We can first use the identity, that $\cosh^2 x - \sinh^2 x = 1$, to simplify

$$g(x) = u(\cosh^2 u - \sinh^2 u) + \pi^u = u + \pi^u$$

Then we have

$$\frac{dg}{du} = \frac{d}{du}u + \pi^u = 1 + \pi^u \ln(\pi)$$